



# AKTU

# B.Tech II-Year



# Cyber security



## Unit-5 in One Shot

## Pdf Notes + PYQs + Imp. Questions

# Unit-5

## AKTU IMP QUESTIONS

| AKTU IMP QUESTIONS |                                                                                           |                    |
|--------------------|-------------------------------------------------------------------------------------------|--------------------|
| 1.                 | Explain different types of security policies that organization should implement           | AKTU 2024-25       |
| 2.                 | Describe the process of developing and implementing a comprehensive cyber security policy | AKTU 2024-25       |
| 3.                 | Explain the need of information security policy in an organisation                        | AKTU 2024-25/23-24 |
| 4.                 | Difference between copyright and patent                                                   | AKTU 2024-25       |
| 5.                 | Give an overview of intellectual property related legislation in India                    | AKTU 2023-24       |
| 6.                 | What is cyber law state some cyber law in India                                           | AKTU 2023-24       |
| 7.                 | What are the common intellectual property issues faced in digital age                     | AKTU 2023-24       |
| 8.                 | What are the penalties and legal consequences for cybercrime in Indian law                | AKTU 2023-24       |
| 9.                 | How are trademark infringements prevented and remedied under Indian law?                  | AKTU 2023-24       |

## What is an Information Security Policy?

An Information Security Policy (ISP) is a **formal document** created by an **organization** that defines **rules, practices, and procedures** to protect its **data, systems, and digital assets** from unauthorized access, misuse, or attacks.

## Why is an Information Security Policy Important?

| Reason                               | Explanation                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------|
| 1. Protects Sensitive Data           | Prevents loss, theft, or unauthorized access to personal and business data. |
| 2. Defines Security Responsibilities | Clearly assigns roles to employees for maintaining data security.           |
| 3. Prevents Cyber Threats            | Helps detect, avoid, and respond to hacking, phishing, or malware attacks.  |
| 4. Ensures Legal Compliance          | Meets data protection laws like <b>IT Act, GDPR, or DPDP Act</b> .          |
| 5. Builds Trust                      | Increases customer and partner confidence in how their data is handled.     |

## Key Components of Information Security

| Component         | Meaning                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------|
| Confidentiality   | Ensures that information is <b>accessible only to authorized users</b> .                  |
| Integrity         | Ensures that data is <b>accurate and unaltered</b> during storage or transfer.            |
| Availability      | Ensures that <b>data and systems are accessible</b> when needed by users.                 |
| Authentication    | Verifying a user's <b>identity</b> (e.g., passwords, biometrics).                         |
| Authorization     | Granting <b>users access</b> based on their roles/permissions.                            |
| Non-repudiation   | Ensuring <b>actions</b> can be <b>traced</b> to a specific user (proof of activity).      |
| Risk Management   | Identifying, evaluating, and addressing security risks.                                   |
| Security Controls | Technical (e.g., firewalls), physical (e.g., locks), and administrative (e.g., policies). |

## Benefits of Information Security Policy

| Benefit                        | Explanation                                                            |
|--------------------------------|------------------------------------------------------------------------|
| 1. Protects Data               | Keeps personal, financial, and business data safe from hackers.        |
| 2. Sets Clear Rules            | Tells employees what they can and cannot do with company information.  |
| 3. Legal Compliance            | Helps the company follow laws (like DPDP Act, GDPR, etc.).             |
| 4. Reduces Risk                | Lowers the chance of cyberattacks, data theft, or system failure.      |
| 5. Guides Incident Response    | Gives a clear plan to follow if there is a cyberattack or data breach. |
| 6. Builds Employee Awareness   | Trains staff to handle data safely and avoid risky behavior.           |
| 7. Protects Company Reputation | Prevents data leaks that could harm trust in the company.              |

## Key Reasons We Need Cyber Law

| Reason                                     | Explanation                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1. To Prevent Cybercrimes                  | Cyber laws help <b>detect, punish, and prevent crimes</b> like hacking, identity theft, and online fraud. |
| 2. To Protect Digital Privacy              | Ensures <b>personal data, financial info, and communications</b> are legally protected.                   |
| 3. To Support E-Commerce & Online Business | Builds <b>trust and security</b> in online transactions and contracts.                                    |
| 4. To Define Digital Rights & Duties       | Clarifies rights and responsibilities of <b>internet users, companies, and service providers</b> .        |
| 5. To Regulate Social Media and Content    | Prevents <b>misuse, defamation, fake news, and hate speech</b> online.                                    |

# Process of Developing and Implementing a Comprehensive Cybersecurity Policy/

## How to Develop and Implement an Information Security Policy

| Step                                 | Activity                            | Description                                                                    |
|--------------------------------------|-------------------------------------|--------------------------------------------------------------------------------|
| 1. Assess Current Environment        | Identify assets, systems, and risks | Know what data needs protection and where the vulnerabilities are.             |
| 2. Define Policy Objectives          | Set clear security goals            | Example: Protect customer data, prevent data breaches, ensure system uptime.   |
| 3. Form a Security Team              | Assign roles and responsibilities   | Include IT staff, legal experts, HR, and senior management.                    |
| 4. Identify Legal & Compliance Needs | Follow relevant laws and standards  | E.g., IT Act 2000, GDPR, ISO 27001, DPDP Act (India).                          |
| 5. Draft the Policy                  | Write rules and procedures          | Cover data protection, access control, acceptable use, incident response, etc. |

# Process of Developing and Implementing a Comprehensive Cybersecurity Policy

| Step                            | Activity                                 | Description                                                                |
|---------------------------------|------------------------------------------|----------------------------------------------------------------------------|
| 6. Review and Approve           | Get feedback and management approval     | Ensure it aligns with business and legal goals.                            |
| 7. Communicate and Train        | Educate employees                        | Train staff on their roles and responsibilities in protecting information. |
| 8. Implement Technical Controls | Deploy security tools and configurations | Firewalls, antivirus, encryption, strong passwords, access control, etc.   |
| 9. Monitor and Audit            | Track compliance and performance         | Regularly check if the policy is followed and effective.                   |
| 10. Update and Improve          | Review periodically                      | Revise policy as technology, threats, and business needs evolve.           |

## Indian cyber law

- Indian Cyber Law refers to the **legal framework** in India that **governs activities** in cyberspace, such as online transactions, data protection, cybercrimes, and electronic communication.
- The Information Technology (IT) Act, 2000  
This is the main cyber law in India.

| Objective of cyber law                 | Explanation                                                                  |
|----------------------------------------|------------------------------------------------------------------------------|
| 1. Legal Recognition of E-transactions | To provide legal status to electronic records and digital signatures.        |
| 2. Prevent Cybercrimes                 | To identify and punish offenses like hacking, phishing, identity theft, etc. |
| 3. Data Security & Privacy             | To ensure protection of personal and sensitive information online.           |
| 4. Regulate E-commerce                 | To support growth of online businesses with legal safeguards.                |
| 5. Establish Authorities               | To create bodies like the Cyber Appellate Tribunal for dispute resolution.   |

# Evolution of Indian Cyber Law to Address Emerging Cyber Threats

| Year/Period | Law/Amendment/Policy                             | Key Features                                                                                      | Purpose/Reason                                                                          |
|-------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 2000        | <b>IT Act, 2000</b> (Information Technology Act) | - First cyber law in India<br>- Legal recognition to e-commerce and digital signatures            | Address growing use of internet, online contracts, and initial cybercrimes              |
| 2008        | <b>IT Amendment Act, 2008</b>                    | - Added cyber terrorism, identity theft, phishing, data privacy- Introduced Section 66A, 66C, 66D | To combat rising cybercrimes like phishing, fake profiles, cyberstalking, and terrorism |
| 2013        | <b>Section 66A struck down (later in 2015)</b>   | - Misuse concerns of 66A- Arrests for social media posts                                          | Strengthen free speech and protect citizens from wrongful arrest                        |
| 2017        | <b>Launch of Digital India Programme</b>         | - Promote safe online services-, Push for cyber hygiene and awareness                             | Increase digital usage with better security and awareness                               |

# Evolution of Indian Cyber Law to Address Emerging Cyber Threats

|         |                                                |                                                                                           |                                                                                                    |
|---------|------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 2018    | National Cyber Security Policy Draft (Revised) | - Framework for protecting critical infrastructure                                        | Improve national-level preparedness and defense against advanced threats                           |
| 2019    | Personal Data Protection Bill (Draft)          | - Rules for data collection, storage, and sharing- Rights to individuals                  | Protect privacy and manage misuse of personal data                                                 |
| 2021    | New IT Rules 2021                              | - Due diligence for social media intermediaries- Grievance redressal- Traceability clause | Control misinformation, hate speech, and ensure user accountability                                |
| 2023    | Digital Personal Data Protection (DPDP) Act    | - Defines personal data rights, Imposes penalties on data misuse, Consent-based data use  | Provide stronger data protection, empower users, and regulate organizations handling personal data |
| Ongoing | Cyber Surakshit Bharat & CERT-In Guidelines    | - Promote cybersecurity best practices ,Guidelines for reporting cyber incidents          | Strengthen cyber defense across sectors, especially for government and critical infrastructure     |

# Rights of Individuals & Organizations under Indian cyber law

| Right                                      | Description                                                                    |
|--------------------------------------------|--------------------------------------------------------------------------------|
| 1. Right to Privacy                        | Protection of personal and sensitive data from unauthorized access.            |
| 2. Right to Legal Recognition of E-records | Electronic records and digital signatures are legally valid.                   |
| 3. Right to Protection from Cybercrime     | Victims can file complaints for cyber offenses like hacking, fraud, stalking.  |
| 4. Right to Data Security                  | Individuals have a right to know how their data is collected and used.         |
| 5. Right to Compensation                   | Victims can claim compensation for damages caused by data theft or negligence. |

## Obligations of Individuals under Indian cyber law

| Obligation                        | Description                                                        |
|-----------------------------------|--------------------------------------------------------------------|
| 1. Avoid Unauthorized Access      | Must not hack or access anyone's data without permission.          |
| 2. Follow Ethical Online Behavior | No cyberbullying, phishing, or spreading fake information.         |
| 3. Report Cybercrimes             | Should report incidents through the Cyber Crime Portal or CERT-In. |
| 4. Respect Digital Signatures     | Should not forge or misuse someone else's digital signature.       |

## Obligations of Organizations under Indian cyber law

| Obligation                                          | Description                                                                   |
|-----------------------------------------------------|-------------------------------------------------------------------------------|
| 1. Data Protection Compliance                       | Must protect users' data and follow SPDI rules.                               |
| 2. Cybersecurity Measures                           | Required to install firewalls, anti-malware, and intrusion detection systems. |
| 3. Incident Reporting                               | Must report major cyber incidents within <b>6 hours</b> of detection.         |
| 4. Content Regulation (Social Media, IT Rules 2021) | Must remove illegal content and appoint grievance officers.                   |
| 5. KYC & User Verification                          | Platforms must verify users' identity as per government rules.                |

## Cybercrimes and Their Penalties

| Cybercrime                                | Law Section                                      | Punishment                                                    |
|-------------------------------------------|--------------------------------------------------|---------------------------------------------------------------|
| <b>Hacking</b>                            | IT Act Sec 66                                    | Jail up to <b>3 years</b> and/or <b>fine up to ₹5 lakh</b>    |
| <b>Identity Theft</b>                     | IT Act Sec 66C                                   | Jail up to <b>3 years</b> and/or <b>fine up to ₹1 lakh</b>    |
| <b>Cheating by impersonation (online)</b> | IT Act Sec 66D                                   | Jail up to <b>3 years</b> and/or <b>fine up to ₹1 lakh</b>    |
| <b>Sending offensive messages</b>         | IT Act Sec 66A<br>( <i>struck down in 2015</i> ) | No longer valid (unconstitutional)                            |
| <b>Cyber terrorism</b>                    | IT Act Sec 66F                                   | <b>Life imprisonment</b>                                      |
| <b>Publishing obscene content online</b>  | IT Act Sec 67                                    | First offence: Jail up to <b>3 years</b> , fine up to ₹5 lakh |

## Cybercrimes and Their Penalties

| Cybercrime                       | Law Section            | Punishment                                                            |
|----------------------------------|------------------------|-----------------------------------------------------------------------|
| Child pornography                | IT Act Sec 67B         | Jail up to <b>5 years</b> and fine up to ₹10 lakh                     |
| Email/Online frauds              | IPC Sec 420 (Cheating) | Jail up to <b>7 years</b> and fine                                    |
| Cyberstalking or harassment      | IPC Sec 354D           | Jail up to <b>3 years</b> (first offence), 5 years for repeat offence |
| Data breach/leak (Personal Data) | DPDP Act, 2023         | Penalty up to <b>₹250 crore</b> depending on severity                 |

## What is Intellectual Property (IP)

It refers to **creations of the human mind**, like inventions, artistic works, brand names, designs, and software, which are **protected by law** so the creator can benefit from their own work

| Type of IP                      | Description                                                                  | Example                         |
|---------------------------------|------------------------------------------------------------------------------|---------------------------------|
| 1. Copyright                    | Protects <b>original literary, artistic, musical</b> , and software works.   | Books, movies, songs, software  |
| 2. Patent                       | Protects <b>new inventions</b> or processes that are useful and non-obvious. | A new medicine or machine       |
| 3. Trademark                    | Protects <b>brand names, logos, slogans</b> , and symbols used in business.  | Nike's "✓" logo, McDonald's "M" |
| 4. Trade Secret                 | Protects <b>confidential business information</b> from competitors.          | Coca-Cola recipe                |
| 5. Industrial Design            | Protects the <b>visual design</b> of objects (shape, color, pattern).        | Car design, mobile phone look   |
| 6. Geographical Indication (GI) | Identifies products from a <b>specific location</b> , known for quality.     | Darjeeling Tea, Banarasi Sarees |

## Common Intellectual Property Issues

| Issue                           | Description                                                                    |
|---------------------------------|--------------------------------------------------------------------------------|
| 1. Copyright Infringement       | Unauthorized copying or sharing of digital books, music, software, videos.     |
| 2. Software Piracy              | Illegal downloading, use, or sale of copyrighted software.                     |
| 3. Trademark Misuse             | Copying logos or brand names to sell fake or lookalike products online.        |
| 4. Domain Name                  | Registering famous brand names as domain names to mislead or sell at profit.   |
| 5. Plagiarism                   | Copying text, graphics, or code without giving credit to the original creator. |
| 6. Data Theft/Trade Secret Leak | Hacking or leaking confidential business data (like algorithms, codes).        |

# Why IP Protection Is Important in the Digital Era

| Reason                         | Explanation                                                                 |
|--------------------------------|-----------------------------------------------------------------------------|
| 1. Easy Copying & Distribution | Digital files can be duplicated and shared globally in seconds.             |
| 2. Protects Innovation         | Creators need legal protection to earn from their inventions or content.    |
| 3. Encourages Creativity       | Strong IP laws motivate artists, developers, and designers to create more.  |
| 4. Prevents Brand Damage       | Protects against fake products that harm brand image and consumer trust.    |
| 5. Supports Economic Growth    | IP-driven industries (films, IT, design) contribute heavily to the economy. |

## Example of an Intellectual Property (IP) Issue

### Case: Software Piracy Faced by Microsoft

#### Situation:

Microsoft, a global software company, often faces **piracy of its Windows operating system and Microsoft Office software** in countries including India.

#### IP Issue Faced:

**Copyright Infringement** – People and organizations **illegally copy and install**

**Microsoft software** without buying proper license

| Effect                          | Description                                                |
|---------------------------------|------------------------------------------------------------|
| <b>Loss of Revenue</b>          | Company loses money when users don't pay for the software. |
| <b>Security Risks to Users</b>  | Pirated software may be bundled with malware.              |
| <b>Legal Action &amp; Fines</b> | Microsoft sues companies using unlicensed copies.          |

## How Businesses and Individuals Can Protect Their Intellectual Property (IP) Rights in the Digital Domain

### 1. Register IP Legally

| Type      | Action                                            |
|-----------|---------------------------------------------------|
| Copyright | Register with the Copyright Office.               |
| Patent    | File a patent application for new inventions.     |
| Trademark | Register logos, names, and slogans with IP India. |
| Design    | Register unique product designs.                  |

### 2. Use Digital Protection Tools

| Tool/Technique                  | Purpose                                               |
|---------------------------------|-------------------------------------------------------|
| Watermarking                    | Prevents unauthorized reuse of images/videos.         |
| Digital Rights Management (DRM) | Controls copying/sharing of digital media.            |
| Encryption                      | Secures confidential designs, code, or trade secrets. |
| Licensing Agreements            | Define how others can legally use your IP.            |

### 3. Monitor and Enforce Rights

| Action                       | Why It Matters                                                     |
|------------------------------|--------------------------------------------------------------------|
| Use IP monitoring tools      | Detect unauthorized use or piracy online.                          |
| File takedown notices (DMCA) | Request platforms like YouTube or Google to remove copied content. |
| Legal action or complaints   | Take violators to court or file with cyber cells/IPR cells.        |

### 4. Educate Employees and Partners

| Steps                    | Benefits                                             |
|--------------------------|------------------------------------------------------|
| Sign NDAs and IP clauses | Protect trade secrets and company inventions.        |
| Train on IP awareness    | Prevent internal misuse or accidental sharing of IP. |

## Intellectual Property (IP) Related Legislation in India

| Legislation                                                      | Purpose                                                  | Covers                                   |
|------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------|
| 1. The Copyright Act, 1957                                       | Protects original creative works.                        | Books, music, films, software, paintings |
| 2. The Patents Act, 1970 (Amended 2005)                          | Grants exclusive rights for inventions.                  | New inventions, processes, and products  |
| 3. The Trade Marks Act, 1999                                     | Protects brand identity.                                 | Logos, brand names, slogans              |
| 4. The Designs Act, 2000                                         | Protects aesthetic design of objects.                    | Product shape, patterns, packaging       |
| 5. The Geographical Indications (GI) Act, 1999                   | Protects products linked to specific geographic origins. | Darjeeling Tea, Banarasi Sarees          |
| 6. The Semiconductor Integrated Circuits Layout-Design Act, 2000 | Protects layout designs of electronic circuits.          | Microchip designs                        |
| 7. The Biological Diversity Act, 2002                            | Protects genetic resources and traditional knowledge.    | Medicinal plants, indigenous practices   |

## Intellectual Property (IP) Related Legislation in India

| Legislation                                                           | Purpose                                           | Covers                    |
|-----------------------------------------------------------------------|---------------------------------------------------|---------------------------|
| <b>8. Protection of Plant Varieties and Farmers' Rights Act, 2001</b> | Protects new plant varieties and farmers' rights. | Agricultural plant breeds |

## What is a Patent?

A **patent** is a **legal right granted by the government** to an inventor for a **new invention**, giving them **exclusive rights** to make, use, sell, or distribute the invention for a certain period (usually **20 years** in India).

### Requirements for Obtaining a Patent in India

To get a patent, an invention must meet the following **criteria** under the **Indian Patents Act, 1970**:

| Requirement                  | Explanation                                                                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Novelty                   | The invention must be <b>new</b> and not known to the public before the filing.                                                                           |
| 2. Inventive Step            | It must involve a <b>technical advancement</b> or improvement                                                                                             |
| 3. Industrial Applicability  | It must be <b>capable of being used</b> in any industry.                                                                                                  |
| 4. Patentable Subject Matter | It should not fall under <b>non-patentable</b> categories (e.g., abstract ideas, laws of nature). It must be a new machine, process, chemical combination |
| 5. Complete Disclosure       | The inventor must clearly explain how the invention works in the application.                                                                             |

## Rights of a Patent Holder in India

A **patent holder** (also called a patentee) enjoys **exclusive legal rights** over their invention under the **Indian Patents Act, 1970**, for a period of **20 years** from the filing date.

| Right                            | Explanation                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| 1. Right to Exclusively Use      | The patentee can <b>make, use, sell, or import</b> the invention.                                 |
| 2. Right to Prevent Misuse       | Can <b>stop others</b> from using, copying, or selling the patented invention without permission. |
| 3. Right to License the Patent   | Can give <b>permission (license)</b> to others to use the invention                               |
| 4. Right to Assign the Patent    | Can <b>transfer ownership</b> (fully or partially) to another person or company.                  |
| 5. Right to Sue for Infringement | Can take <b>legal action</b> if someone uses the invention without permission.                    |
| 6. Right to Exploit Commercially | Can earn <b>royalties or profits</b> by using or licensing the invention.                         |

## How Individuals or Businesses Can Apply for and Protect Their Patent in India

Applying for and protecting a patent involves a **legal process** managed by the **Indian Patent Office** under the **Patents Act, 1970**.

| Step                         | Details                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| 1. Check Patentability       | Ensure the invention is <b>new, useful, and non-obvious</b> (meets all criteria).                      |
| 2. Draft Patent Application  | Prepare <b>provisional or complete specification</b> with detailed description, drawings, and claims.  |
| 3. File Application with IPO | Submit online or offline to the <b>Indian Patent Office</b>                                            |
| 4. Publication               | Application is published in the patent journal <b>after 18 months</b> (can request early publication). |
| 5. Request for Examination   | File a request for examination (Form 18) within <b>48 months</b> of filing.                            |
| 6. Response to Objections    | If examiner raises objections, <b>respond with clarifications</b> or amendments.                       |
| 7. Grant of Patent           | If accepted, patent is <b>granted and published</b> in the patent journal.                             |

## What is Copyright?

**Copyright** is a **legal right** granted to the **creator of original works** (like books, music, software, films, etc.), giving them **exclusive control** over how their work is used, copied, or distributed.

## Requirements for Obtaining Copyright in India

Unlike patents, **copyright is automatic**—you get it as soon as you **create an original work** in a tangible form.

However, **registration is recommended** to prove ownership in case of dispute

| Requirement               | Explanation                                                                |
|---------------------------|----------------------------------------------------------------------------|
| 1. Originality            | The work must be <b>original</b> and not copied from another source.       |
| 2. Fixed in Tangible Form | It must be <b>written, recorded, painted, or typed</b> (not just an idea). |
| 3. Creative Expression    | The work must involve some <b>creativity or intellectual effort</b> .      |

## Rights of a Copyright Holder in India

Under the **Copyright Act, 1957**, the creator or owner of a copyrighted work is given **exclusive rights** to control the use and distribution of their work.

### Main Rights of a Copyright Holder

| Right                          | Description                                                                    |
|--------------------------------|--------------------------------------------------------------------------------|
| 1. Right to Reproduce          | Only the copyright holder can make <b>copies</b> of the work.                  |
| 2. Right to Distribute         | Right to <b>sell, rent, or publish</b> the work to the public.                 |
| 3. Right to Public Performance | Allows performing the work publicly (e.g., playing music or screening a film). |

### Duration of Rights

| Type of Work                            | Protection Duration                                   |
|-----------------------------------------|-------------------------------------------------------|
| Literary, musical, artistic works       | <b>Lifetime of the author + 60 years</b> after death. |
| Cinematographic films, sound recordings | <b>60 years</b> from publication year.                |

## Steps to Register Copyright in India.

| Step                    | Description                                                                              |
|-------------------------|------------------------------------------------------------------------------------------|
| 1. Prepare the Work     | Ensure the work is <b>original and in tangible form</b> (written, recorded, etc.).       |
| 2. Apply Online         | Visit the official site: <a href="https://copyright.gov.in">https://copyright.gov.in</a> |
| 3. Fill Form XIV        | Provide details of the creator, nature of work, and ownership.                           |
| 4. Submit the Work      | Upload or send a <b>copy of the work</b> (manuscript, video, audio, code, etc.).         |
| 5. Pay the Fee          | Fees vary by type (literary, artistic, musical, etc.).                                   |
| 6. Wait for Examination | The Copyright Office reviews and publishes the application.                              |
| 7. Certificate Issued   | If no objections, a <b>registration certificate</b> is granted.                          |

## How to Protect Copyright After Registration

| Protection Method           | Purpose                                                                |
|-----------------------------|------------------------------------------------------------------------|
| Display Copyright Notice    | Use © symbol with name and year (e.g., © 2025 ABC Corp).               |
| Use Licensing Agreements    | Define how others may legally use the work.                            |
| Digital Watermarking/DRM    | Prevent unauthorized copying or distribution online.                   |
| Monitor for Infringement    | Regularly check online platforms or marketplaces for unauthorized use. |
| Take Legal Action if Needed | Send legal notices or file a case in court against infringers.         |

## What is a Trademark?

A trademark is a **unique symbol, word, name, logo, or combination** used by a business to **identify its goods or services** and distinguish them from others.

## Requirements for Registering a Trademark in India

| Requirement                 | Explanation                                                                     |
|-----------------------------|---------------------------------------------------------------------------------|
| 1. Distinctiveness          | It must clearly <b>identify your brand</b> and not be generic or commonly used. |
| 2. Not Deceptive            | Should not <b>mislead</b> the public about the product or service.              |
| 3. Not Identical or Similar | Must not be too <b>similar to an existing registered trademark</b> .            |
| 4. Lawful and Ethical       | Cannot include offensive, obscene, or prohibited symbols                        |
| 5. Used or Proposed to Use  | You must either be <b>using it already</b> or <b>intend to use it soon</b> .    |

## Rights of a Trademark Holder in India

| Right                              | Description                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------|
| 1. Exclusive Right to Use          | Only the holder can use the trademark for the goods or services it is registered under. |
| 2. Right to License or Assign      | The holder can <b>license</b> the trademark to others or <b>transfer ownership</b> .    |
| 3. Right to Sue for Infringement   | Can take legal action against anyone who uses the mark <b>without permission</b> .      |
| 4. Right to Use ® Symbol           | After registration, can use ® to show it's legally protected.                           |
| 5. Right to Protect Brand Identity | Prevent others from using <b>confusingly similar</b> names, logos, or packaging.        |
| 6. Right to Earn Revenue           | Can <b>earn money through licensing</b> or franchising the trademark.                   |

## Steps to Apply for a Trademark in India for organisation / individual

| Step                                | Description                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1. Trademark Search                 | Check for existing similar trademarks at <a href="https://ipindia.gov.in">https://ipindia.gov.in</a> . |
| 2. Choose Correct Class             | Select the correct <b>trademark class</b> (1 to 45) based on goods/services.                           |
| 3. File Application (Form TM-A)     | Submit online through the IP India portal or offline.                                                  |
| 4. Pay Government Fee               | ₹4,500 (individual/startup) or ₹9,000 (company) per class (subject to change).                         |
| 5. Examination by Registrar         | Trademark Office checks for any objections or conflicts.                                               |
| 6. Publication in Trademark Journal | If accepted, published to invite public objections (for 4 months).                                     |
| 7. Registration Certificate         | If no opposition, certificate is issued — valid for <b>10 years</b> , renewable.                       |

# How to Protect a Registered Trademark

| Protection Method        | Why It Matters                                                 |
|--------------------------|----------------------------------------------------------------|
| Use ® Symbol             | Shows the mark is legally protected.                           |
| Renew Every 10 Years     | File renewal to <b>keep the protection active</b> .            |
| Monitor for Infringement | Watch for unauthorized or similar use of your mark.            |
| Send Legal Notices       | Warn infringers with <b>cease-and-desist letters</b> .         |
| File Infringement Suits  | Take legal action for damages and to stop misuse.              |
| Use Proper Branding      | Consistently use your trademark in the same design and format. |

## Explain different types of security policies that organization should implement

Organizations need multiple **types of security policies** to protect data, systems, networks, and users from internal and external cyber threats.

### 1. Information Security Policy

**Purpose:** Sets the foundation for protecting all digital and physical data.

**Includes:** Data classification, access control, confidentiality, and compliance.

### 2. Acceptable Use Policy (AUP)

**Purpose:** Defines how employees can use company devices, internet, and email.

**Includes:** No personal use of office systems, banning illegal downloads, social media rules.

### 3. Access Control Policy

**Purpose:** Ensures only authorized users can access certain data or systems.

**Includes:** Role-based access, password rules, login permissions, and account lockout policies.

### 4. Data Protection and Privacy Policy

**Purpose:** Safeguards personal and sensitive data from misuse or breach.

**Includes:** Data handling rules, encryption, retention, and compliance (e.g., DPDP Act, GDPR).

## 5. Network Security Policy

### Purpose:

- Protects the organization's IT network infrastructure.
- **Includes:** Firewall rules, VPN access, Wi-Fi usage, intrusion detection systems.

## 6. Remote Access Policy

- **Purpose:** Manages secure access to systems by remote employees.
- **Includes:** VPN usage, device authentication, and remote monitoring.

## 7. Incident Response Policy

- **Purpose:** Guides staff on how to respond to a cyberattack or data breach.
- **Includes:** Reporting steps, response teams,, and recovery procedures.

## What is the DPDP Act (Digital Personal Data Protection Act), 2023?

The Digital Personal Data Protection (DPDP) Act, 2023 is a law passed by the Indian government to regulate the **collection, storage, and processing of personal data** in digital form.

### Main objective DPD act

| Objective                         | Explanation                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------|
| 1. Protect Personal Data          | Safeguards digital personal data from unauthorized access, misuse, or leakage.                 |
| 2. Empower Data Principals        | Gives individuals control over their data — right to consent, access, and correction.          |
| 3. Ensure Fair Data Processing    | Organizations must use data <b>lawfully, fairly, and for defined purposes</b> .                |
| 4. Promote Transparency           | Data fiduciaries must inform users how their data is used, stored, and shared.                 |
| 5. Establish Regulatory Authority | Sets up a <b>Data Protection Board of India</b> to enforce the law and handle complaints.      |
| 6. Enable Cross-Border Data Flow  | Allows <b>secure transfer of personal data to certain countries</b> , supporting global trade. |

# Scope of the Digital Personal Data Protection (DPDP) Act, 2023

| Scope Area                             | Explanation                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Applies to Personal Data</b>        | Data that can <b>identify a person</b> , like name, address, Aadhaar number, phone, etc.       |
| <b>Covers Digital Data Only</b>        | Applies to <b>data collected online</b> or later digitized (e.g., scanned forms).              |
| <b>Applies Across India</b>            | Applies to <b>all individuals and organizations</b> working in India, including government.    |
| <b>Also applies internationally</b>    | Covers foreign companies <b>processing Indian citizens' data</b> , if offering goods/services. |
| <b>Both Private &amp; Govt. bodies</b> | Applies to <b>private companies, start-ups, apps, and government departments</b> .             |

# Rights Given to Individuals under the Digital Personal Data Protection (DPDP) Act, 2023

| Right                           | Explanation                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------|
| 1. Right to Access Information  | You can <b>ask what data is collected</b> , how it's used, and who it's shared with.                |
| 2. Right to Consent             | Your <b>consent is required</b> before your personal data is collected or used.                     |
| 3. Right to Withdraw Consent    | You can <b>withdraw your consent</b> at any time, and the processing must stop.                     |
| 4. Right to Correction          | You can <b>correct, update, or complete</b> your personal data if it's inaccurate.                  |
| 5. Right to Erasure             | You can <b>ask for deletion</b> of your data when it's no longer needed or if consent is withdrawn. |
| 6. Right to Grievance Redressal | You can <b>file a complaint</b> with the Data Fiduciary or the <b>Data Protection Board</b> .       |
| 7. Right to Nominate            | You can <b>nominate someone</b> to exercise your rights in case of death or incapacity.             |

## Obligations Imposed on Organizations (Data Fiduciaries) by the DPDP Act, 2023

| Obligation                       | Explanation                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1. Obtain Valid Consent          | Must collect personal data <b>only after clear, informed, and specific consent</b> from the user.                |
| 2. Provide Notice                | Must <b>inform the user</b> about the purpose of data collection, how it will be used, and their rights.         |
| 3. Limit Purpose of Use          | Data must be used <b>only for the purpose stated</b> at the time of collection.                                  |
| 4. Data Minimization             | Should collect <b>only the minimum data</b> necessary to fulfill the intended purpose.                           |
| 5. Ensure Accuracy               | Must ensure that personal data is <b>correct, complete, and up to date</b> .                                     |
| 6. Erase Data on Withdrawal      | Must <b>delete personal data</b> when consent is withdrawn or when data is no longer needed.                     |
| 7. Implement Security Safeguards | Must use <b>technical and organizational measures</b> to protect data from unauthorized access, loss, or misuse. |

# Legal Framework Governing Cyber Activities in India

| Component                                                   | Description / Purpose                                                                                            |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1. Information Technology (IT) Act, 2000                    | The <b>primary cyber law</b> in India. Covers electronic governance, cybercrime, and data security.              |
| 2. IT Rules (Amendments: 2011, 2021, 2022)                  | Includes rules for <b>intermediaries</b> , social media platforms, and handling user data.                       |
| 3. Digital Personal Data Protection (DPDP) Act, 2023        | Ensures <b>privacy and protection</b> of personal data; gives rights to individuals and duties to organizations. |
| 4. Indian Telegraph Act, 1885                               | Regulates <b>telecommunication infrastructure</b> , including surveillance and monitoring.                       |
| 5. Indian Penal Code (IPC) & Bharatiya Nyaya Sanhita (2023) | Used to <b>punish cybercrime offences</b> like identity theft, cyberstalking, and fraud.                         |
| 6. Companies Act, 2013                                      | Mandates companies to ensure <b>cyber risk management</b> and report data breaches.                              |
| 7. Judicial Oversight                                       | Indian courts play a role in <b>interpreting cyber laws</b> and ensuring justice in cybercrime cases.            |

## Latest development in Indian cyber law

| Category                  | Development                                         | Purpose / Impact                                                                                                                                                           |
|---------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New Laws & Policy Updates | DPDP Rules                                          | DPDP Act is being put into action by enforcing rules like <b>getting user consent</b> before collecting data and <b>storing or deleting data</b> securely and responsibly. |
|                           | IT Rules Amendments                                 | Addresses <b>AI-generated deepfakes</b> , stricter intermediary obligations                                                                                                |
|                           | Proposed New IT Act                                 | Broader recognition of <b>e-signatures</b> , tougher on cybercrimes                                                                                                        |
| Criminal Law Reforms      | Bharatiya Nyaya Sanhita, 2023 (Effective July 2024) | Recognizes <b>cybercrime as organized crime</b> ; digital records admissible in court                                                                                      |
| Institutional Reforms     | e-Zero FIR (Delhi Pilot)                            | Allows <b>online FIR filing</b> for cyber frauds                                                                                                                           |
|                           | I4C under PMLA, 2002                                | Enables data-sharing for financial cybercrime enforcement                                                                                                                  |
|                           | CBI's Bharatpol Portal (2025)                       | Real-time <b>cross-border cybercrime intelligence sharing</b>                                                                                                              |
|                           | India–US Cyber MoU                                  | <b>Joint cybercrime investigation and info-sharing</b> with FBI                                                                                                            |

# Latest development in Indian cyber law

| Category                  | Development                                   | Purpose / Impact                                                                                                                                             |
|---------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI & Surveillance Tools   | Cyber Suraksha Online (Maharashtra)           | Uses AI to detect crimes against women and children                                                                                                          |
|                           | Deepfake Detection Tools (VastavX AI)         | Deployed in Karnataka and other states                                                                                                                       |
|                           | 24/7 Social Media Monitoring (Kolkata Police) | Detects hate speech, fake news, sensitive content                                                                                                            |
| Telecom & Messaging Rules | New Telecom Security Rules (Nov 2024)         | Mandates <b>24/7 security centers</b> , breach reporting in 6 hrs                                                                                            |
|                           | TRAI Bulk SMS Traceability                    | Tracks phishing, spam, and financial fraud via messaging                                                                                                     |
| Enforcement Practices     | Operation Ghost SIM                           | Police or cyber agencies have <b>caught gangs</b> that were using <b>fake or illegally issued SIM cards</b>                                                  |
|                           | Rajasthan Cyber Guidelines                    | Defines clear rules for freezing accounts in cyber fraud cases                                                                                               |
|                           | Telangana Cyber Slavery Crackdown             | Authorities are investigating <b>human trafficking gangs</b> that are <b>sending people to other countries</b> and <b>forcing them to work in cybercrime</b> |

## AKTU IMP QUESTIONS

|    |                                                                                           |                    |
|----|-------------------------------------------------------------------------------------------|--------------------|
| 1. | Explain different types of security policies that organization should implement           | AKTU 2024-25       |
| 2. | Describe the process of developing and implementing a comprehensive cyber security policy | AKTU 2024-25       |
| 3. | Explain the need of information security policy in an organisation                        | AKTU 2024-25/23-24 |
| 4. | Difference between copyright and patent                                                   | AKTU 2024-25       |
| 5. | Give an overview of intellectual property related legislation in India                    | AKTU 2023-24       |
| 6. | What is cyber law state some cyber law in India                                           | AKTU 2023-24       |
| 7. | What are the common intellectual property issues faced in digital age                     | AKTU 2023-24       |
| 8. | What are the penalties and legal consequences for cybercrime in Indian law                | AKTU 2023-24       |
| 9. | How are trademark infringements prevented and remedied under Indian law?                  | AKTU 2023-24       |

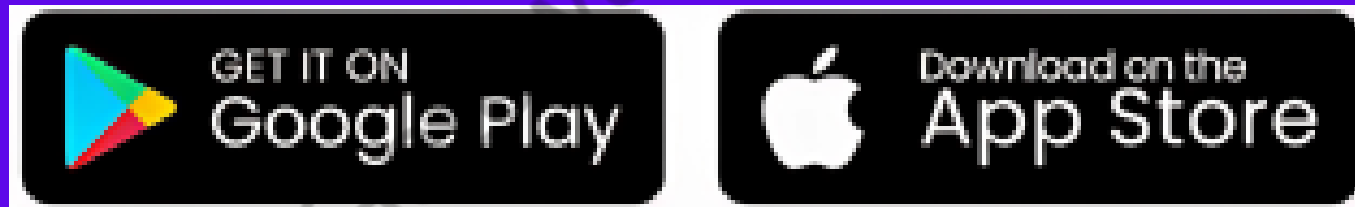


# Gateway Classes

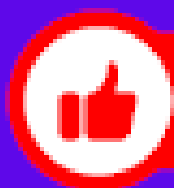
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# Thank You

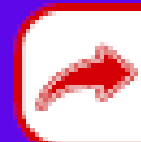


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